

OSSP (Operating Systems And Systems Programming)

Course code: 25CS2104E

Practical-02: system calls open(), read(), write(), and close()

Code Output:

```
sushma_sree@LAPTOP-6PJSRT08:~$ nano practical2.c
sushma_sree@LAPTOP-6PJSRT08:~$ gcc practical2.c
sushma_sree@LAPTOP-6PJSRT08:~$ ./a.out
Source file descriptor: 3
Destination file descriptor: 4
File copied successfully..
```

Contents of source file:

```
sushma_sree@LAPTOP-6PJSRT08:~$ cat sourcefile
hello
this is sushma
from kl university
this is my os assignment
sushma_sree@LAPTOP-6PJSRT08:~$ |
```

Contents of destination file:

```
sushma_sree@LAPTOP-6PJSRT08:~$ cat destinationfile
hello
this is sushma
from kl university
this is my os assignment
sushma_sree@LAPTOP-6PJSRT08:~$ |
```

B)

```
sushma_sree@LAPTOP-6PJSRT08:~$ strace cat b1
execve("/usr/bin/cat", ["cat", "b1"], 0x7ffe5079e438 /* 25 vars */) = 0
brk(NULL) = 0x557ab75ff000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x79c10938b000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20087, ...}) = 0
mmap(NULL, 20087, PROT_READ, MAP_PRIVATE, 3, 0) = 0x79c109386000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x79c109353000
mmap(0x79c10935a000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x79c10935a000
mmap(0x79c10937b000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x79c10937b000
mmap(0x79c109382000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x79c109382000
mmap(0x79c109384000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x79c109384000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x79c109325000
mmap(0x79c109329000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x79c109329000
mmap(0x79c10934d000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x79c10934d000
mmap(0x79c109351000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x79c109351000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
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fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0
mmap(NULL, 1200152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x79c1091ff000
mmap(0x79c109211000, 655360, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x12000) = 0x79c109211000
mmap(0x79c1092b1000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xb2000) = 0x79c1092b1000
mmap(0x79c109323000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x123000) = 0x79c109323000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\3\0>\0\1\0\0\0 \250\2\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 840, 64) = 840
fstat(3, {st_mode=S_IFREG|0755, st_size=2186512, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 840, 64) = 840
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x79c108e00000
mmap(0x79c108e28000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x79c108e28000
mmap(0x79c108fc0000, 319488, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1c0000) = 0x79c108fc0000
mmap(0x79c10900e000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20d000) = 0x79c10900e000
mmap(0x79c109014000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x79c109014000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libcres2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=699056, ...}) = 0
mmap(NULL, 701112, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x79c109153000
mmap(0x79c109156000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x3000) = 0x79c109156000
mmap(0x79c1091d2000, 176128, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7f000) = 0x79c1091d2000
mmap(0x79c1091fd000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xa9000) = 0x79c1091fd000
close(3) = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x79c109151000
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mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x79c10914e000
arch_prctl(ARCH_SET_FS, 0x79c10914eb00) = 0
set_tid_address(0x79c10914f128) = 2222
set_robust_list(0x79c10914ede0, 24) = 0
rseq(0x79c10914e6a0, 0x20, 0, 0x53053053) = 0
mprotect(0x79c10900e000, 16384, PROT_READ) = 0
mprotect(0x79c1091fd000, 4096, PROT_READ) = 0
mprotect(0x79c109323000, 4096, PROT_READ) = 0
mprotect(0x79c109351000, 4096, PROT_READ) = 0
mprotect(0x79c109382000, 4096, PROT_READ) = 0
mprotect(0x557aa0df2000, 1462272, PROT_READ) = 0
mprotect(0x79c1093d0000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
getrandom("\xca\xfa\x45\x26\x01\x34\xa2", 8, GRND_NONBLOCK) = 8
munmap(0x79c109386000, 20087) = 0
statfs("/sys/fs/selinux", {f_type=SYSFS_MAGIC, f_bsize=4096, f_blocks=0, f_bfree=0, f_bavail=0, f_files=0, f_ffr
ee=0, f_fsid={val=[0x5248cee9, 0x39c42646]}, f_namelen=255, f_frsize=4096, f_flags=ST_VALID|ST_NOSUID|ST_NODEV|S
T_NOEXEC|ST_NOATIME}) = 0
statfs("/selinux", 0x7ffd1150c130) = -1 ENOENT (No such file or directory)
brk(NULL) = 0x557ab75ff000
brk(0x557ab7620000) = 0x557ab7620000
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbdf...", 1024) = 442
lseek(3, 0, SEEK_CUR) = 442
lseek(3, 416, SEEK_SET) = 416
close(3) = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "none /usr/lib/modules/6.18.33.2-...", 1024) = 1024
read(3, "e 0 \ndepts /dev/pts devpts rw,", 1024) = 1024
read(3, "osuid,nodev,noexec,relatime,nosy", 1024) = 1024

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read(3, "/user/1000 tmpfs rw,nosuid,nodev...", 1024) = 102
read(3, "", 1024) = 0
close(3) = 0
access("/etc/selinux/config", F_OK) = -1 ENOENT (No such file or directory)
fcntl(0, F_GETFD) = 0
fcntl(1, F_GETFD) = 0
fcntl(2, F_GETFD) = 0
rt_sigaction(SIGPIPE, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
poll([{fd=0, events=0}, {fd=1, events=0}, {fd=2, events=0}], 3, 0) = 0 (Timeout)
rt_sigaction(SIGPIPE, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x79c108
e45cb0}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "557aa0489000-557aa057b000 r--p 0"... , 1024) = 1024
read(3, "0 08:30 13628", 1024) = 1024
read(3, "00 08:30 13631", 1024) = 1024
read(3, " /usr/lib/x86_64-lin", 1024) = 1024
read(3, "0 13625 /us", 1024) = 566
close(3) = 0
sched_getaffinity(2222, 32, [0 1 2 3 4 5 6 7]) = 8
rt_sigaction(SIGSEGV, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
sigaltstack(NULL, {ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=0}) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x79c109388000
mprotect(0x79c109388000, 4096, PROT_NONE) = 0
sigaltstack({ss_sp=0x79c109389000, ss_flags=0, ss_size=8192}, NULL) = 0
gettid() = 2222
rt_sigaction(SIGSEGV, {sa_handler=0x557aa07ae0c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_res
torer=0x79c108e45cb0}, NULL, 8) = 0
rt_sigaction(SIGBUS, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=0x557aa07ae0c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_rest
orer=0x79c108e45cb0}, NULL, 8) = 0

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prctl(PR_GET_AUXV, 0x7ffd1150bc20, 0x200, 0, 0) = 448
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/uucore", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
statx(3, "", AT_STATX_SYNC_AS_STAT|AT_EMPTY_PATH, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=4080, ...}) = 0
read(3, "# Common strings shared across a"... , 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x79c108e45cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x79c108e45cb0}, 8) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=~[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x79c108e45cb0}, {sa_handler=0x557aa07ae0c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x79c108e45cb0}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=~[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x79c108e45cb0}, {sa_handler=0x557aa07ae0c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x79c108e45cb0}, 8) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOK|ECHOE|ECHO|ECHOCTL|ECHOKE, ...}) = 0
brk(0x557ab7641000) = 0x557ab7641000
statx(AT_FDCWD, "b1", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=55, ...}) = 0
openat(AT_FDCWD, "b1", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=55, ...}) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0

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splice(3, NULL, 1, NULL, 1048576, 0) = -1 EINVAL (Invalid argument)
pipe([4, 5]) = 0
fcntl(4, F_SETPIPE_SZ, 1048576) = 1048576
splice(3, NULL, 5, NULL, 1048576, 0) = 55
splice(4, NULL, 1, NULL, 55, 0hello
this is sushma
os class
this is cse
this is klh
) = 55
splice(3, NULL, 5, NULL, 1048576, 0) = 0
close(5) = 0
close(4) = 0
close(3) = 0
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
munmap(0x79c109388000, 12288) = 0
exit_group(0) = ?
+++ exited with 0 +++
sushma_sree@LAPTOP-6PJSRT08:~$ |

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